AI & COMPUTER-VISION BASED SURFACE & PRODUCT DEFECT INSPECTION

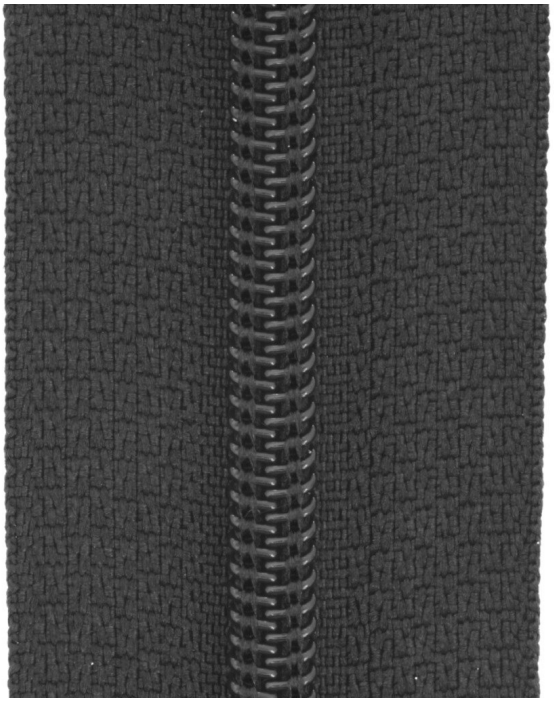
Automated Quality Inspection Report

Inspection ID:	DEF-2026-0317	**Date & Time:**	2026-11-02 13:13:19
Product / Part:	Unnamed Product	**Batch / Lot:**	-
Product Line:	METAL SHEET	**Station:**	Line 1 · Final QC
Inspector:	Line Supervisor	**Outcome:**	**REVIEW**

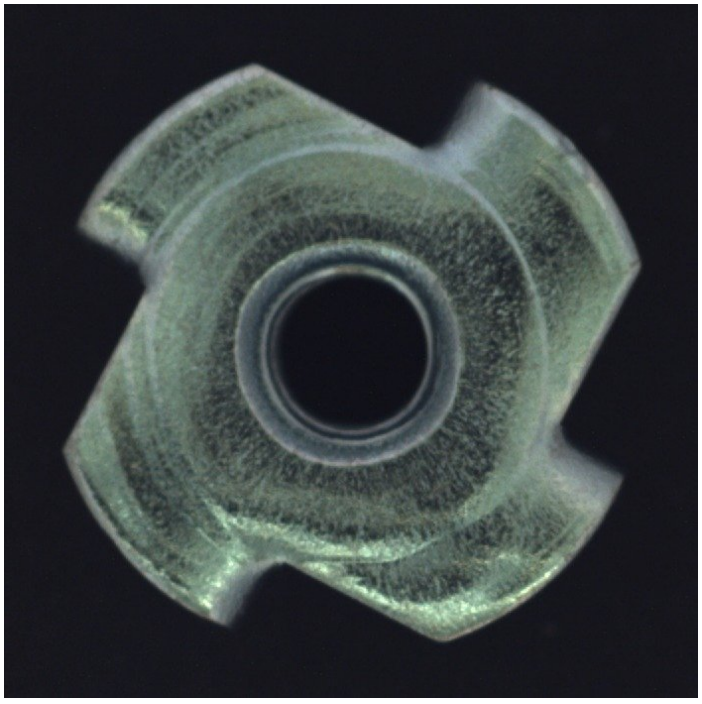
DEFECT SUMMARY

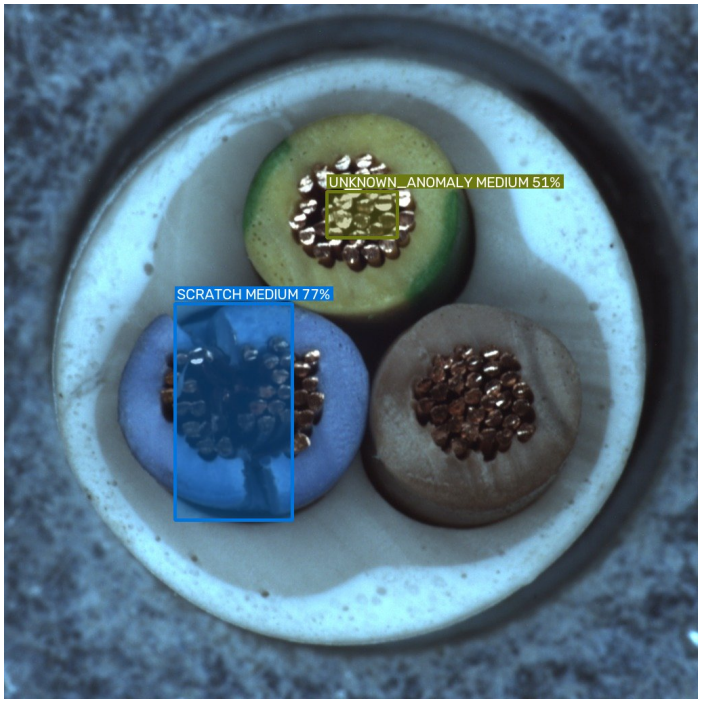
Defect Type	Count	Critical	High	Medium	Low
Scratch	1	0	0	1	0
Unknown Anomaly	1	0	0	1	0

ANNOTATED VISUAL EVIDENCE









DETECTION LOG

#	Type	Severity	Confidence	Location (px)	Detector	Review
1	Scratch	**MEDIUM**	77%	252,441 - 424,759	YOLO_SEG +ANOMALY	PENDING
2	Unknown Anomaly	**MEDIUM**	51%	476,276 - 580,344	ANOMALY	PENDING

ROOT-CAUSE HINTS

Scratch (1)
• Abrasive contact during conveyor transfer or manual handling
• Burrs or debris on guide rails, fixtures, or tooling
• Insufficient protective film / interleaving between stacked parts
→ Inspect and deburr guide rails, grippers, and transfer chutes
→ Add protective film or separators at stacking and packing stations

Unknown Anomaly (1)
• Process variation not covered by the built-in knowledge base
→ Collect confirmed samples and perform a 5-Why / fishbone analysis with the line team

DETERMINATION & SIGN-OFF

Inspector Remarks: Defect inspection reviewed and signed off by quality inspector

Analysed by: Defect Inspection Engine (YOLO_SEG+ANOMALY)

Quality Sign-off: _____